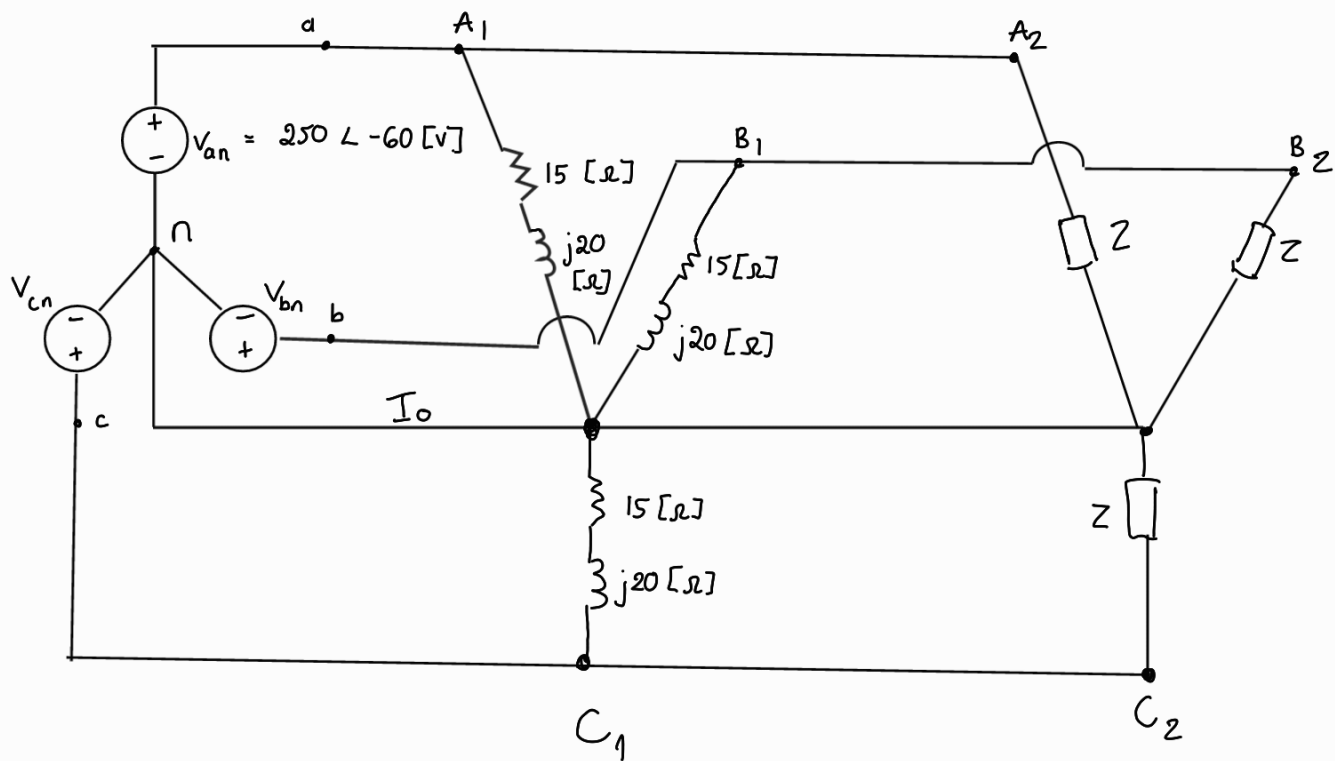


a)



b.) $S_{a2} = 500 \angle 45^\circ \text{ [VA]}$

$V_{an} = 250 \angle -60^\circ \text{ [V]}$

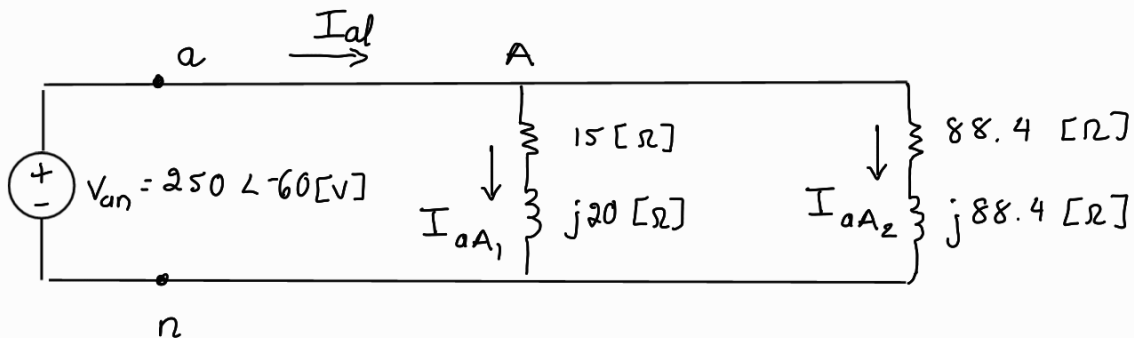
Note;

$$S = V_{rms} \cdot I_{rms}^+$$

$$\Rightarrow Z_2^* = \frac{|V_{an}|^2}{S} = 125 \angle -45^\circ \text{ } [\Omega]$$

$$\Rightarrow Z_2 = 125 \angle 45^\circ \text{ } [\Omega] = 88.4 + j88.4 \text{ } [\Omega]$$

a-phase equivalent circuit:



$$c) |I_{aA1}| = \left| \frac{V_{an}}{Z_{load1}} \right| = \frac{250}{|15 + j20|} = \frac{250}{25} = 10 \text{ [A]}$$

$$d) |V_{A_n}| = |V_{an}| = 250 \text{ [V]}$$

$$e) I_{al} = I_{aA_1} + I_{aA_2} = \frac{250 \angle -60^\circ}{15 + j20} + \frac{250 \angle -60^\circ}{88.4 + j88.4}$$

$$= 10 \angle -113.23^\circ \text{ [A]}$$

$$\Rightarrow |I_{al}| = 10 \text{ [A]}$$

$$f) S_{del. by. Total source} = 3 (S_{abs. by. load1} + S_{abs. by. load2})$$

$$S_{abs. by. load1} = \frac{|V_{an}|^2}{Z_{load1}^*} = \frac{250^2}{15 - j20} = 1500 + 2000j \text{ [VA]}$$

$$S_{abs. by. load2} = \frac{|V_{an}|^2}{Z_{load2}^*} = \frac{250^2}{88.4 - j88.4} = 353.51 + 353.51j \text{ [VA]}$$

$$\Rightarrow S_{del. by. Total source} = 5560.52 + 7060.52j \text{ [VA]}$$

$$= 8987.23 \angle 51.78^\circ \text{ [VA]}$$

$$g) S_{del. by. Total Source} = \underset{\substack{\text{average} \\ \text{Power}}}{P} + j \underset{\substack{\text{reactive} \\ \text{power}}}{Q}$$

$$\Rightarrow \begin{cases} P = 5.56 \text{ [kW]} \\ Q = 7.06 \text{ [kVAR]} \end{cases}$$

$$h) \begin{cases} pf = \cos(51.8^\circ) = 0.618 \\ Q > 0 \Rightarrow \text{lagging} \end{cases}$$

i). power factor use $\cos(\theta)$ and $\cos(\theta) = \cos(-\theta)$

so the numerical value of power factor does not indicate whether the current leads or lags the voltage

- Reactive power uses $Q = |S| \sin \theta$, whose sign already indicates inductive or capacitive behavior
 - $\left\{ \begin{array}{l} Q < 0 : \text{capacitive or leading current} \\ Q > 0 : \text{inductive or lagging current} \end{array} \right.$